Econ 580: Honors Tutorial in Research Project Design

Instructors:

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Course Description

The class is meeting weekly on Tuesdays and Thursdays, 11am - 12:15pm, in SOC SCI 6112. You can ignore the time and location for the final exam in the registrar's course information.

Students will be required to complete a project that requires original research in economics.

Additional Information

This course will introduce you to the many facets and phases of completing a research paper in economics. As you work toward the ultimate goal of an original research project of your own design, we will help you master both its content and delivery (including the critical and often overlooked skills of effective oral and written communication). The resulting research paper can then be used as the foundation for an honors thesis, or you could start afresh: there is a lot of learning about how to do research that takes place regardless of whether you continue your specific research project as your thesis.

Prerequisites

You must meet department requirements to take this course. Please consult department staff about whether you meet these requirements.

Credits

This 3-credit course has two 75 minute lectures per week. Students are expected to work approximately 6 hours per week outside class to complete assignments and learn the relevant material.

Learning Outcomes

At the end of the course students will be able to:

A. read, understand, and critically evaluate existing research in economics;
B. conduct original research in economics, including developing economic models and analyzing primary data sources;
C. present their original research and defend their methods and conclusions;
D. write a complete and well-structured economics research paper.

Assignments

All written materials must be double-spaced, have 1-inch margins on all sides, and use 12-point font. Page lengths do not include tables, figures, or reference lists.

You can use any program you would like to create your paper drafts (for example Microsoft Word, Pages, or LaTeX). No matter how you produce your documents, all written materials must be submitted as PDF files.

If your work includes mathematical fonts, Greek letters, or other non-standard characters, you may want to use the LaTeX typesetting system. It is the standard in economics and If you are planning to go to graduate school you will learn it sooner or later anyway. This being said, the use of LaTeX is recommended, but not required. There is a substantial start-up cost to using LaTeX and you should not attempt to use it if this will reduce your research effort substantially.

You can create LaTeX documents using Overleaf (https://www.overleaf.com/edu/wisc), Scientific Word/Workplace, LyX (http://www.lyx.org), or by directly typing LaTeX code, which is then typeset with software like TeXShop (on a Mac computer) or WinEdt (on a Windows machine). We have tried all of these but found Overleaf to be one of the most user-friendly options. We also recommend the book *LaTeX: A Document Preparation System* by Leslie Lamport.

Below is an overview of the assignments you are required to complete in this course. These descriptions are somewhat sparse and generic. The actual assignments with more detailed instructions will be available on the Canvas course site.

Assignment 1: Introduction to Version Control

Git is widely used in the software industry and is rapidly becoming the norm for collaborative work in economics and academia more broadly. The goal of this assignment is to familiarize yourself with Git and GitHub by completing a step-by-step version control tutorial from Software Carpentry. This will get you set up for the remainder of the semester, as you will document all of the steps of your independent work on GitHub.

Assignment 2: Replication Exercise and Write-Up

Replication exercises are a useful learning-by-doing way to get started with research. This is likely to involve using methods and software packages that you have had very little experience with, and you will need to work more independently to troubleshoot issues that arise (as they inevitably do in research!). The goal of this exercise is to expose you to a more independent, real-world computational experience and instill in you the confidence to carry out your own computational work in your research project.

For this assignment, you will choose one of a handful of papers we have pre-selected to replicate and modestly extend. You will download the data used in the paper, write code to analyze the data, create tables and/or figures of results, and include a write-up.

Your write-up should be 5 pages and provide an overview of the paper, a discussion of your attempt to replicate the results and any issues you encountered, your results from the replication, and a modest extension to the paper. This should be written as a paper (not an outline), and include proper citations. It should be written clearly in your own words and without grammatical errors, as clear exposition is an important part of the research process.

You will also attach your programming code to your GitHub project and include a README file for others to be able to replicate your work.

Assignment 3: First Draft

At this point you have started to think about your own research project. For this assignment, submit a 1-page document with your research question and the motivation behind it. Be sure to include at least five relevant sources (such as peer-reviewed articles or working papers).

If you haven't settled on a single topic yet it is perfectly fine to submit two or more potential research questions. For each proposed research question (if there are more than one), the write-up should be at least half a page and include at least five relevant sources.

Submit a PDF file to the corresponding Canvas assignment. We will post all drafts in the "Download student drafts" module on Canvas within 24 hours after the submission deadline.

Assignment 4: Peer Review of First Draft

An important part of the research process is receiving feedback on drafts of your work at various stages and incorporating this feedback into the next draft. This means that you should also be well-versed in providing feedback on other researchers' work.

In this assignment, you will write two one-page peer reviews of the first drafts of two other students (randomly assigned). Your review should restate the research question and motivation (in your own words), and provide suggestions for improvement. These suggestions can include ways to reframe the question or motivation, ideas for other data sources or empirical strategies, tips on exposition, or anything else you think would be helpful for the student.

Make sure that these suggestions are constructive and not mean-spirited. All peer reviews will be posted on discussion boards on Canvas for the class to see and comment on.

Assignment 5: Second Draft

At this point you should have finalized your research question. For this assignment, expand on the first draft by making any edits to your research question and motivation based on the evolution of your research idea and feedback from the class, start fleshing out a literature review, and include a discussion of the proposed methods, data, and models, if appropriate. This draft should be at least 3 pages and should include a list of all of your references in a common citation format.

Submit a PDF file to the corresponding Canvas assignment. We will post all drafts in the "Download student drafts" module on Canvas within 24 hours after the submission deadline.

Assignment 6: Peer Review of Second Draft

Write two one-page peer reviews of the second drafts of two other students (newly randomly assigned). Your review should restate the research question and motivation (in your own words), and provide suggestions for improvement. These suggestions can include ways to reframe the question or motivation, ideas for other data sources or empirical strategies, tips on exposition, or anything else you think would be helpful for the student.

Make sure that these suggestions are constructive and not mean-spirited. All peer reviews will be posted on discussion boards on Canvas for the class to see and comment on.

Assignment 7: Third Draft

Revise and expand on your second draft based on your progress and feedback from the class. In addition, this draft should include a title page with an abstract (150 words or less) as well as an initial summary of the empirical and/or theoretical results that you have generated so far. Discuss your results, justify the choice of the data you used in your analysis, and motivate the methods you selected. If your work is based on a mathematical model, discuss why it is salient for the question you are trying to answer.

This draft should be at least 10 pages (not including the title page).

Submit a PDF file to the corresponding Canvas assignment. We will post all drafts in the "Download student drafts" module on Canvas within 24 hours after the submission deadline.

Assignment 8: Peer Review of Third Draft

Write two one-page peer reviews of the second drafts of two other students (newly randomly assigned). Your review should restate the research question and motivation (in your own words), and provide suggestions for improvement. These suggestions can include ways to reframe the question or motivation, ideas for other data sources or empirical strategies, tips on exposition, or anything else you think would be helpful for the student.

Make sure that these suggestions are constructive and not mean-spirited. All peer reviews will be posted on discussion boards on Canvas for the class to see and comment on.

Assignment 9: In-Class Presentation

Present the final results of your work in a 15-minute presentation that meets academic standards in the last two weeks of the semester. This presentation should reflect the elements in your third draft and include a brief introduction to your research question and its motivation, a description of the methodology used to address the question, and a clear summary of your key findings. It is important to explain the implications of your results, how they connect to existing literature, and what they contribute to the field. During the presentation, you should be prepared to answer questions about your approach and defend your assumptions. We strongly encourage you to meet with us beforehand to discuss your slides and progress to ensure your presentation is effective and insightful for the class.

Assignment 10: Final Research Paper

The final version of your semester project should be at least 25 pages in length and must have the following elements:

1. a title;
2. an abstract (no more than 150 words);
3. an introduction;

4. a literature review;
5. a description of the model, if you used it all;
6. a description of your data and your empirical methods;
7. a discussion of your empirical results and potential robustness checks;
8. a conclusion.

Present your empirical results using tables and figures as you see fit. Use your references to get ideas for an effective presentation and illustration of your results.

Assignment 11: 5-Minute Progress Report Presentations

Over the semester, you will deliver three 5-minute Progress Report Presentations. The following guidelines outline the expectations for each of these presentations.

Progress Report Presentation #1: This presentation should reflect the elements you will include in your first draft: You will present your research question and the motivation behind it. Clearly articulate the question you aim to answer, provide relevant background information, and explain why this question is important.

Progress Report Presentation #2: This presentation should reflect the elements you will include in your second draft: You will briefly present your research question and motivation, and you will connect your project with the relevant literature you are trying to contribute to. This presentation should emphasize what has been done in the literature and what is new in your project relative to the cited work, so make sure to devote most of the 5 minutes to this.

Progress Report Presentation #3: This presentation should reflect the elements you will include in your third draft, focusing primarily on the empirical and/or theoretical results you have generated so far. Begin with a brief reminder of your research question and motivation, skipping the literature review for this presentation. Use one slide to present the data used in your analysis and another to explain the methods used. If your work involves a mathematical model,discuss its relevance to the question you are trying to answer. Conclude by presenting your results, comparing them to previous studies, and discussing their implications.

Assignment 12: Active Participation Throughout Semester

Class participation:

We expect you to attend all in-class meetings and provide constructive and respectful feedback during classmates' in-class progress reports and final presentations.

1-on-1 meetings with instructors:

During the semester, you will be required to meet with us to discuss the progress of your project. It is important to come prepared and take this part of the course seriously. The meetings are brief, lasting no more than 15 minutes, so making the best use of this time is essential to maximize the value of our interactions. How can you ensure productive and effective meetings?

Start by identifying the specific questions or issues you want to discuss. Think ahead, create a list of these questions, and prioritize them. To help us prepare for the meeting, please send this list to us the night before via email.

If you intend to share results during the meeting, prepare a concise set of slides that summarize your work—such as the regressions you are running, an outline of your model, or key findings—and highlight the aspects you would like to discuss. Be sure to email these slides to the instructor you are meeting with at least one day in advance to ensure a focused and productive discussion.

Evaluation and Grades

Assignment 1: Introduction to version control	3%
Assignment 2: Replication exercise and referee report	10%
Assignment 3: First draft	5%
Assignment 4: Peer review of first draft	2%
Assignment 5: Second draft	5%
Assignment 6: Peer review of second draft	2%
Assignment 7: Third draft	10%
Assignment 8: Peer review of third draft	4%
Assignment 9: Presentation	10%
Assignment 10: Final document	30%
Assignment 11: 5-minute progress report presentations	9%
Assignment 12: Class participation and 1-on-1 meetings	10%

Resources

Career Counseling

If you would like to learn more about career and internship opportunities for economics students or discuss the career skills you learn as an economics major you are encouraged to meet with

an economics career advisor by emailing econcareers@ssc.wisc.edu or visiting the Economics Career Development Office's website: econ.wisc.edu/careers/.

University Resources

1. UW-Madison Writing Center: http://writing.wisc.edu/
2. Social Science Computing: https://www.ssc.wisc.edu/sscc/instruction/
3. Seminars in Economics:
    a. https://econ.wisc.edu/workshop-calendar/
    b. http://www.irp.wisc.edu/index.htm
    c. https://bus.wisc.edu/
    d. http://www.aae.wisc.edu/
    e. https://www.lafollette.wisc.edu/

References for Writing and Presenting in Economics

Dudenhefer, *A Guide to Writing in Economics*

Cochrane, *Writing Tips for PhD Students*

McCloskey, *Economical Writing*

McCloskey, *The Rhetoric of Economics*

Schwabish, *Better Presentations: A Guide for Scholars, Researchers, and Wonks*

Strunk and White, *Elements of Style*

Reference for Econometrics

Angrist and Pischke, *Mastering Metrics*

Course Schedule (Subject to Revision)

Date	Week		Assignments (due on Friday that week)
1/20/26	1	Introduction, version control, replication (part 1)	
1/22/26		Introduction, version control, replication (part 2)	
1/27/26	2	Econometrics review	
1/29/26		Examples of previous papers	Assignment 1 (Introduction to Version Control)

2/3/26	3	Individual 15-minute meetings	
2/5/26		Individual 15-minute meetings	
2/10/26	4	Presentation of replication results	
2/12/26		Presentation of replication results	Assignment 2 (Replication Exercise)
2/17/26	5	Individual 15-minute meetings	
2/19/26		Individual 15-minute meetings	
2/24/26	6	In-class 5-minute progress report 1	
2/26/26		In-class 5-minute progress report 1	Assignment 3 (First Draft)
3/3/26	7	Individual 15-minute meetings	
3/5/26		Individual 15-minute meetings	Assignment 4 (Peer Review of First Draft)
3/10/26	8	In-class 5-minute progress report 2	
3/12/26		In-class 5-minute progress report 2	Assignment 5 (Second Draft)
3/17/26	9	Individual 15-minute meetings	
3/19/26		Individual 15-minute meetings	Assignment 6 (Peer Review of Second Draft)
3/24/26	10	In-class 5-minute progress report 3	
3/26/26		In-class 5-minute progress report 3	
3/31/26	11	Spring break	
4/2/26			
4/7/26	12	Individual 15-minute meetings	
4/9/26		Individual 15-minute meetings	Assignment 7 (Third Draft)
4/14/26	13	Individual 15-minute meetings	
4/16/26		Individual 15-minute meetings	Assignment 8 (Peer Review of Third Draft)
4/21/26	14	Final presentations	Assignment 9 (Presentation)
4/23/26		Final presentations	
4/28/26	15	Final presentations	
4/30/26		Final presentations	
			Assignment 10 (Final Document)

Diversity & Inclusion

Diversity is a source of strength, creativity, and innovation for UW-Madison. We value the contributions of each person and respect the profound ways their identity, culture, background, experience, status, abilities, and opinion enrich the university community. We commit ourselves to the pursuit of excellence in teaching, research, outreach, and diversity as inextricably linked

goals. The University of Wisconsin-Madison fulfills its public mission by creating a welcoming and inclusive community for people from every background – people who as students, faculty, and staff serve Wisconsin and the world.

Department Commitment to Diversity, Equity, and Inclusion

We are committed to making the Department of Economics a welcoming and supportive environment for students, staff, and faculty. To enact this goal, we recognize and value the unique contributions of a diverse community with respect to race, ethnicity, sexual orientation, gender identity, disabilities, sex, age, country of origin, religious beliefs, and differences in socioeconomic, cultural and educational backgrounds. For more information, see https://econ.wisc.edu/dei-wiecon-home/.

Academic Integrity

By virtue of enrollment, each student agrees to uphold the high academic standards of the University of Wisconsin-Madison; academic misconduct is behavior that negatively impacts the integrity of the institution. Cheating, fabrication, plagiarism, unauthorized collaboration, and helping others commit these previously listed acts are examples of misconduct which may result in disciplinary action. Examples of disciplinary action include, but are not limited to, failure on the assignment/course, written reprimand, disciplinary probation, suspension, or expulsion.

Use of Generative AI

In this course, the use of generative AI tools like ChatGPT is permitted if and only if:

1. it is properly acknowledged and attributed and
2. it follows the guidelines below.

If you are not sure if a particular use is permitted, seek clarification from the course instructors. Clarification requests and approvals are only valid if they are in writing.

When you use generative AT tools, follow the guidelines below.

- You may use generative AI as a tool to brainstorm, to find examples, to clarify your ideas, to iterate or refine your answers, or to improve your writing. This type of use is similar to working collaboratively with another student, which is encouraged on certain assignments in this course. You must document your interactions by saving your prompts and the corresponding material generated by the "bot". We have the right to request this documentation and failure to make it available is considered an instance of academic misconduct, irrespective of whether the omission is accidental or deliberate. Moreover, if you end up including content produced by generative AI in your work,

whether it be verbatim or by paraphrasing, you must cite it accordingly. Follow a standard citation style when you do so (e.g., APA, Chicago,...).

- Plugging homework questions into ChatGPT and submitting the output as *your own work* is academically dishonest and is not permitted. If we have reason to believe that you submitted content produced by a bot without proper attribution you will not receive credit for your work and we will reserve the right to initiate a formal academic misconduct process. Beware that we do not have to prove beyond a reasonable doubt that you engaged in this type of academic misconduct. If you cannot credibly demonstrate that you submitted your own work, we will determine that your conduct was academically dishonest.
- In general, be aware that generative AI does not “know” anything. It tries to statistically predict how a reasonably qualified person would respond to the question you submit to the bot. AI can generate inaccurate facts and answers that sound plausible but are factually wrong! Ethan Mollick, a professor at Wharton, has the following note of caution regarding the use of generative AI: “Don't trust anything it says. If it gives you a number or fact, assume it is wrong unless you either know the answer or can check in with another source. It works best for topics you understand.”

Lastly, but most importantly, keep in mind that if you are relying on generative AI to do all the work for you, you may eventually be replaced by it. Use your college education to acquire the kinds of skills that give you access to jobs that stay one step ahead of AI!

Accommodations for Students with Disabilities

The University of Wisconsin-Madison supports the right of all enrolled students to a full and equal educational opportunity. The Americans with Disabilities Act (ADA), Wisconsin State Statute (36.12), and UW-Madison policy (Faculty Document 1071) require that students with disabilities be reasonably accommodated in instruction and campus life. Reasonable accommodations for students with disabilities is a shared faculty and student responsibility. Students are expected to inform faculty [me] of their need for instructional accommodations by the end of the third week of the semester, or as soon as possible after a disability has been incurred or recognized. Faculty [I], will work either directly with the student [you] or in coordination with the McBurney Center to identify and provide reasonable instructional accommodations. Disability information, including instructional accommodations as part of a student’s educational record, is confidential and protected under FERPA.

Academic Misconduct

Academic Integrity is critical to maintaining fair and knowledge based learning at UW Madison. Academic dishonesty is a serious violation: it undermines the bonds of trust and honesty between members of our academic community, degrades the value of your degree and defrauds those who may eventually depend upon your knowledge and integrity.

Examples of academic misconduct include, but are not limited to: cheating on an examination (copying from another student's paper, referring to materials on the exam other than those explicitly permitted, continuing to work on an exam after the time has expired, turning in an exam for regrading after making changes to the exam), copying the homework of someone else, submitting for credit work done by someone else, stealing examinations or course materials, tampering with the grade records or with another student's work, or knowingly and intentionally assisting another student in any of the above. Students are reminded that online sources, including anonymous or unattributed ones like Wikipedia, still need to be cited like any other source; and copying from any source without attribution is considered plagiarism.

The Department of Economics will deal with these offenses harshly following UWS14 procedures:

1. The penalty for misconduct in most cases will be removal from the course and a failing grade.
2. The department will inform the Dean of Students as required and additional sanctions may be applied.
3. The department will keep an internal record of misconduct incidents. This information will be made available to teaching faculty writing recommendation letters and to admission offices of the School of Business and Engineering.

If you think you see incidents of misconduct, you should tell your instructor about them, in which case they will take appropriate action and protect your identity. You can also choose to contact our department administrator, Tammy Herbst-Koel (therbst@wisc.edu), and your identity will be kept confidential.

For more information, refer to https://conduct.students.wisc.edu/academic-misconduct/

Grievance Procedure

The Department of Economics grievance procedures can be found here: https://econ.wisc.edu/undergraduate/grievances/

Mental Health and Violence Prevention Resources

As a student you may experience a range of issues that can cause barriers to learning. These might include strained relationships, anxiety, high levels of stress, alcohol/drug problems, feeling down, or loss of motivation. University Health Services is here to help with these or other issues you may experience. You can learn about the free, confidential mental health services available on campus by calling 608-265-5600 or visiting uhs.wisc.edu. Help is always available.

Experiences of sexual assault, sexual harassment, dating violence, domestic violence, and stalking affect many students on this campus and can create barriers to learning. UW-Madison is committed to fostering a safe, productive learning environment and offers a variety of resources and options for students impacted by violence. Learn about the free, confidential services available on campus and in the community by visiting www.uhs.wisc.edu/assault.

Course Evaluations

Students will be provided with an opportunity to evaluate their enrolled courses and their learning experience. Most instructors use HelioCampus, a digital course evaluation survey tool. In most instances, students receive an official email two weeks prior to the end of the semester, notifying them that anonymous course evaluations are available. Student participation is an integral component of course development, and confidential feedback is important. UW-Madison strongly encourages student participation in course evaluations.